

Business Process Technologies and Management

1. Introduction

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Overview

- Introduction
- Conceptual Process Modeling
- Process Verification
- Process Execution and Automation
- Interorganizational Processes
- Process Monitoring and Logging
- Summary and Outlook

Teaching objectives

This course shall

- develop your *analytical skills* in respect to the analysis, design and modeling of business processes
- give you insights into *fundamental concepts* of process-aware information systems
- enable you to create executable process models
- develop your *skills* in implementing process-aware applications
- *increase your knowledge* about current trends in BPM research and BPM tools

Literature

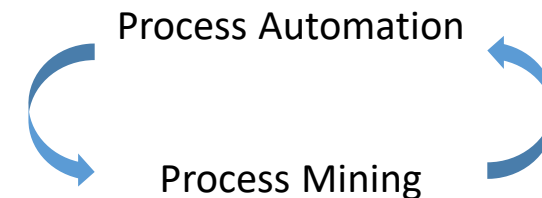


- Weske, M. (2019) Business Process Management - Concepts, Languages, Architectures, Third Edition. Springer 2019, ISBN 978-3-662-59431-5, pp. 1-417 2019
- Marlon Dumas, Marcello La Rosa, Jan Mendling, Hajo A. Reijers: Fundamentals of Business Process Management, Second Edition. Springer 2018, ISBN 978-3-662-56508-7, pp. 1-527
- Wil M. P. van der Aalst: Process Mining - Data Science in Action, Second Edition. Springer 2016, ISBN 978-3-662-49850-7, pp. 3-452
- Wilfried Grossmann, Stefanie Rinderle-Ma: Fundamentals of Business Intelligence. Data-Centric Systems and Applications, Springer 2015, ISBN 978-3-662-46530-1, pp. 1-327
- Manfred Reichert, Barbara Weber: Enabling Flexibility in Process-Aware Information Systems - Challenges, Methods, Technologies. Springer 2012, ISBN 978-3-642-30408-8, pp. I-XVIII, 1-515

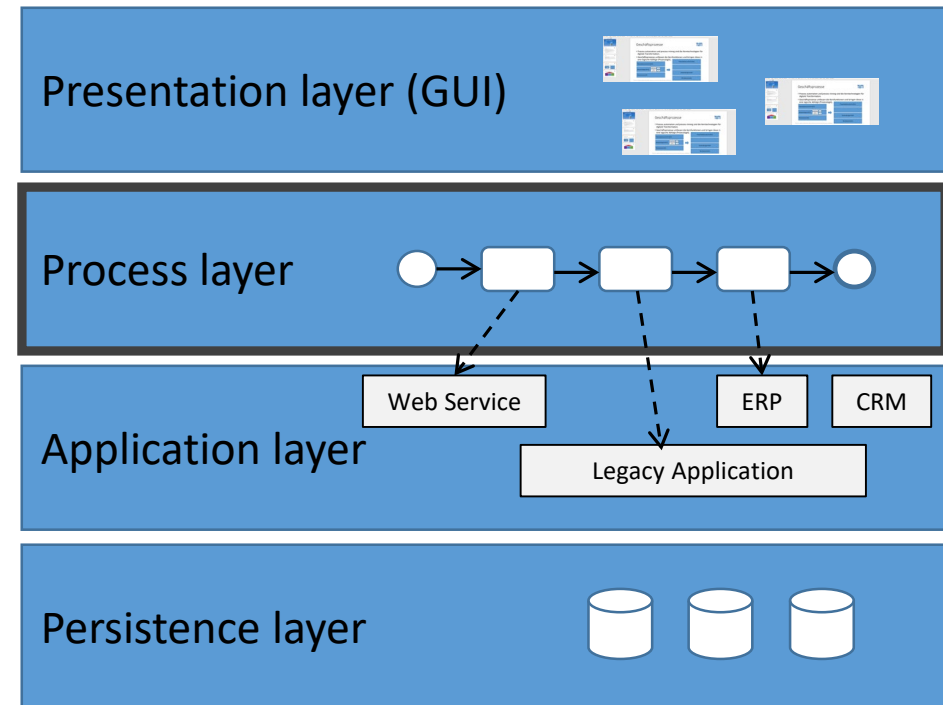
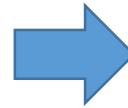
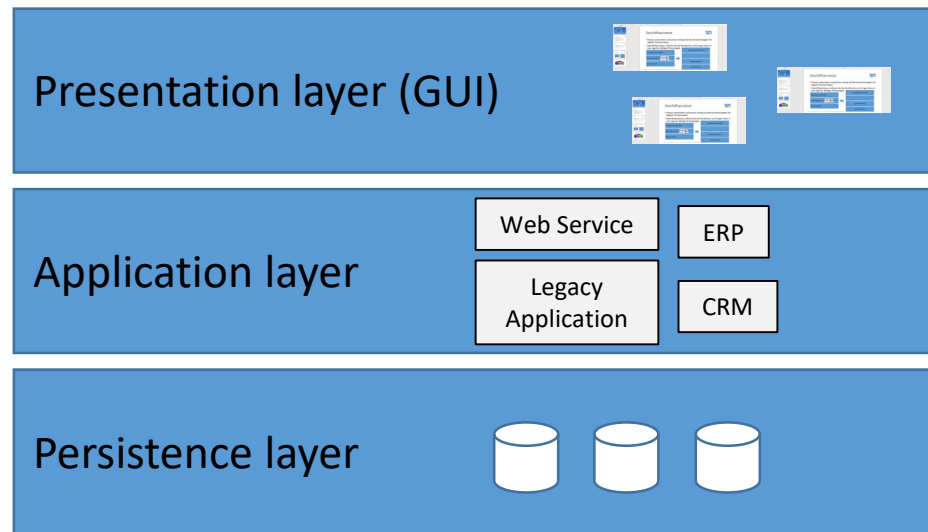
→ more literature provided in the chapters

Motivation

- Business process technologies and management enable
 - documentation
 - optimization
 - Process automation
 - separation of process logic and application code
 - application integration
 - contextualized data collection
 - Process mining:
 - improves transparency
 - provides insights
- key technologies for digital transformation
- independent of the application domain

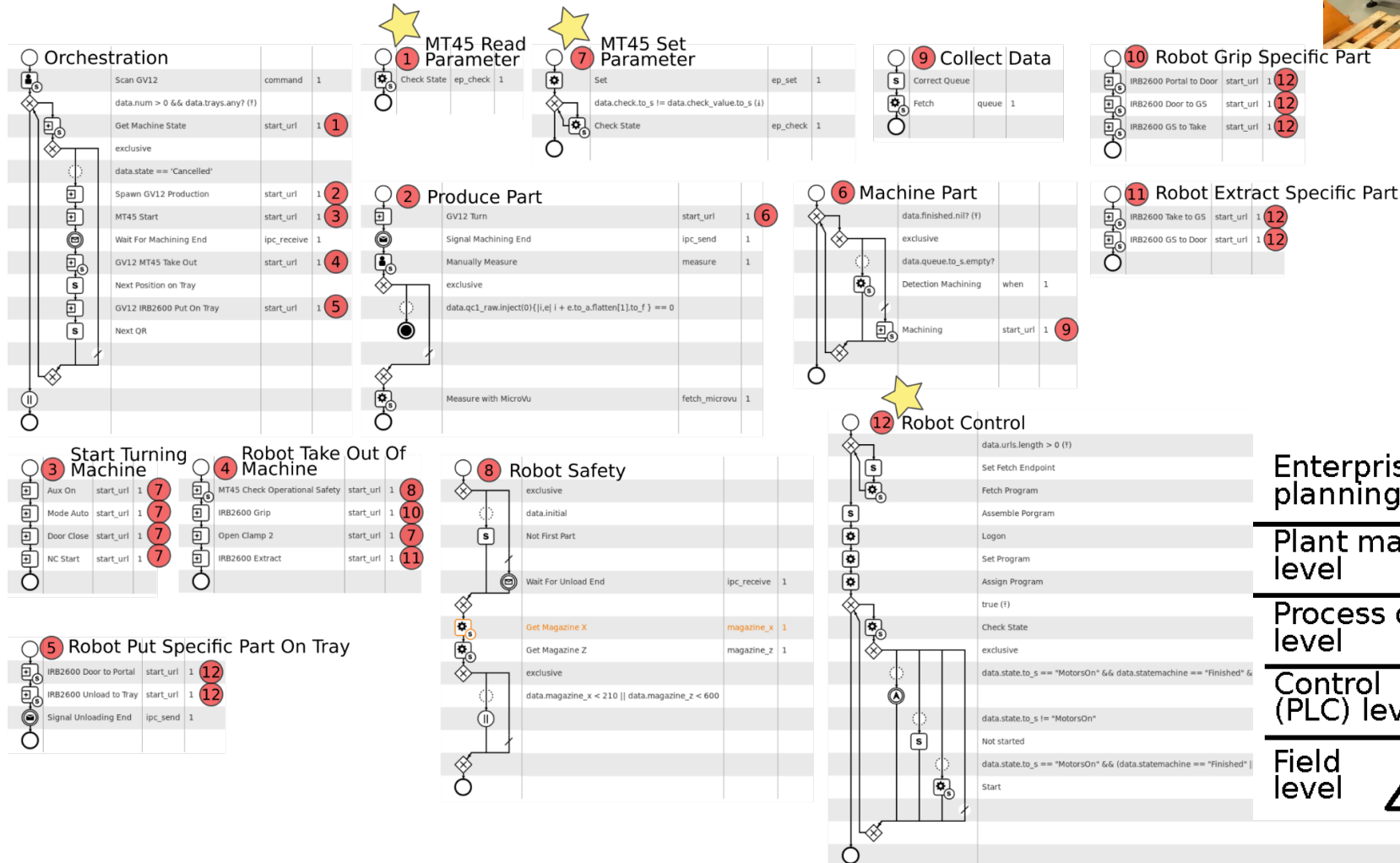


Enterprise architecture



Let us look at one of the most challenging application domains...

Manufacturing processes



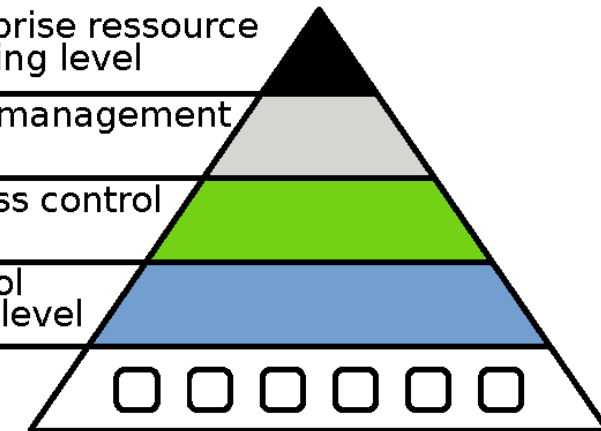
Enterprise resource planning level

Plant management level

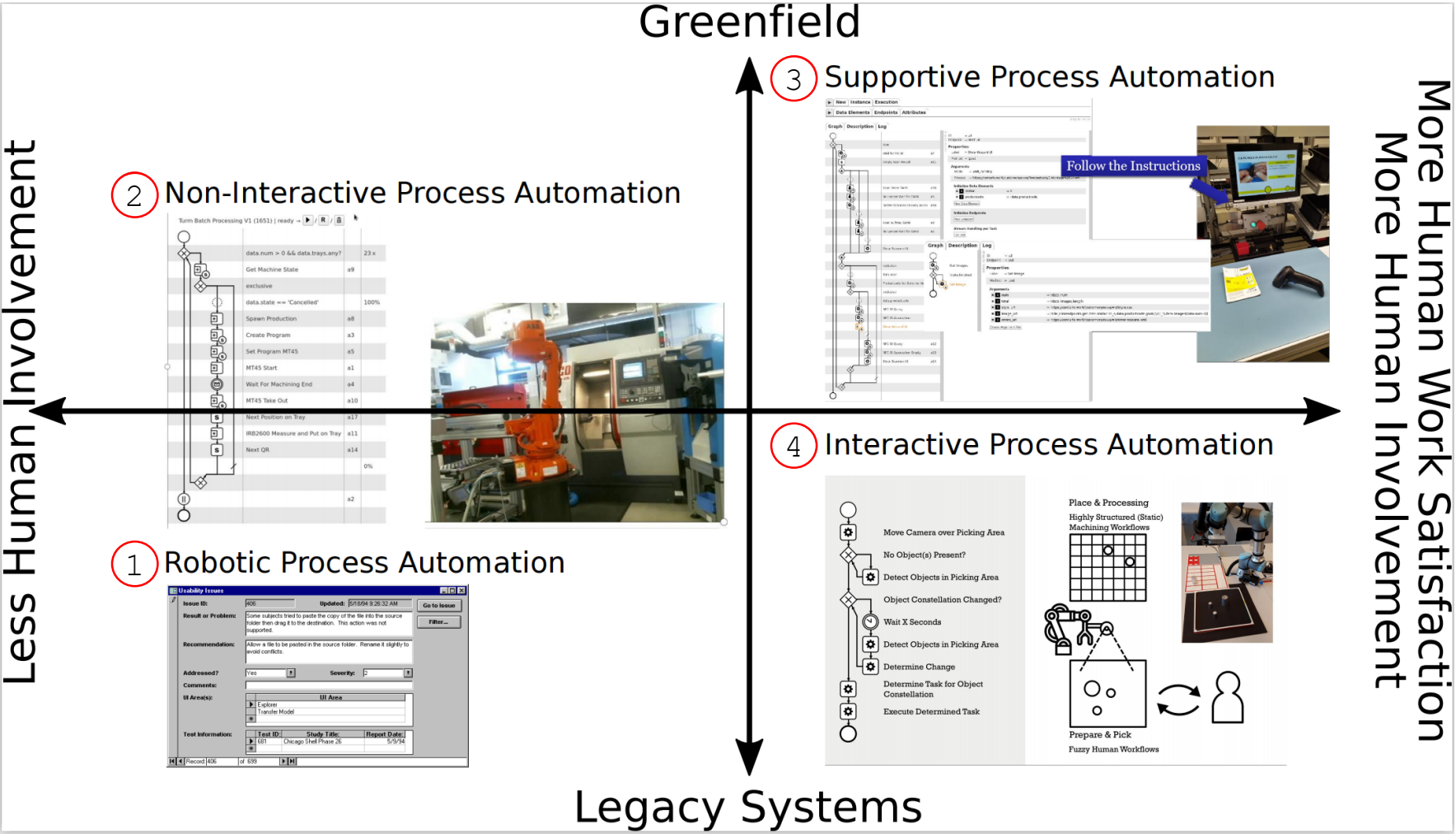
Process control level

Control (PLC) level

Field level



Process automation scenarios

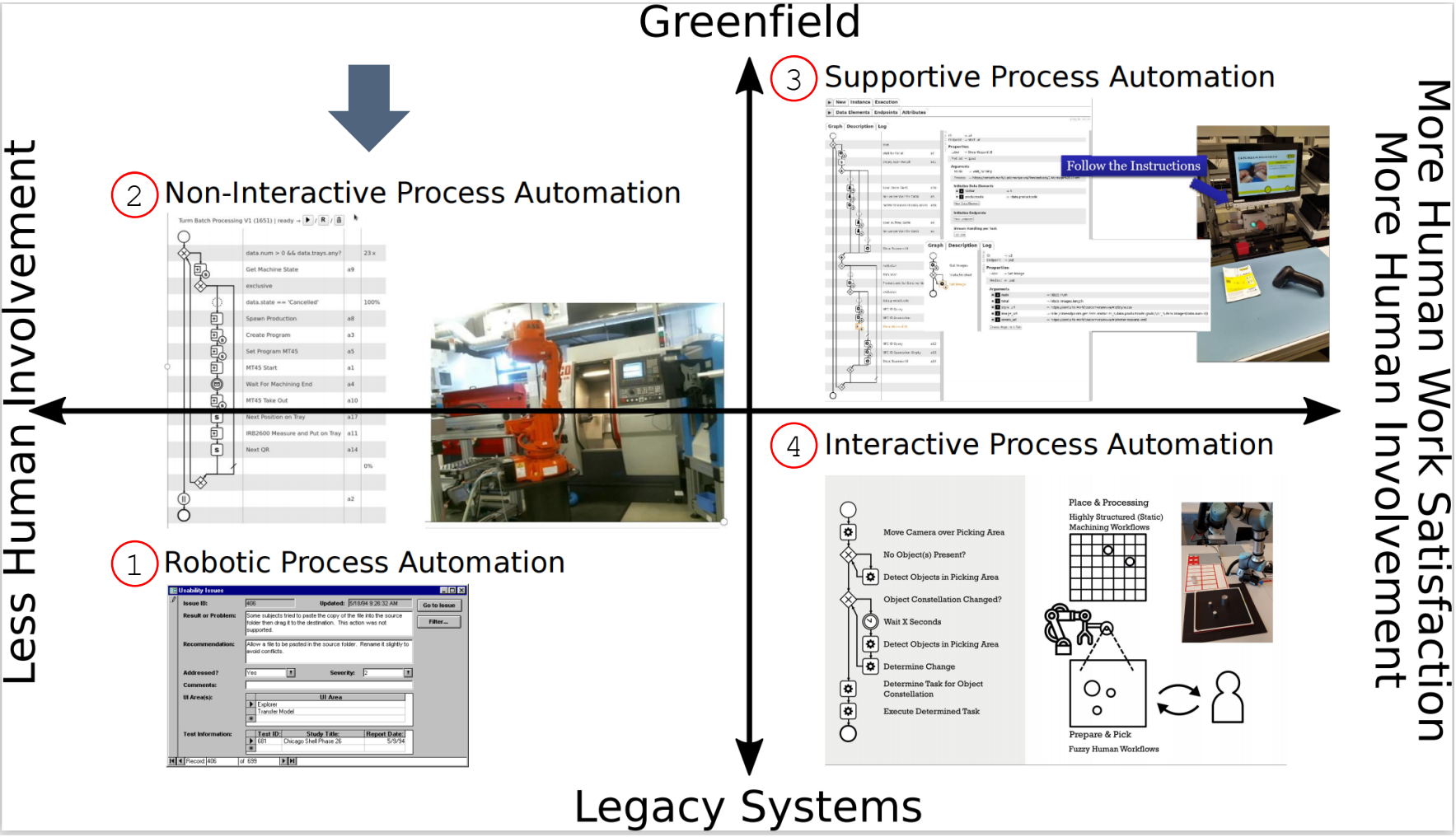


Experience



- Almost two decades of research
- Manufacturing process orchestration engine centurio.work based on cpee.org
- Currently 16 manufacturing process scenarios have been realized across 7 companies
- Benefits: traceability, flexibility, sustainability

Process automation scenarios

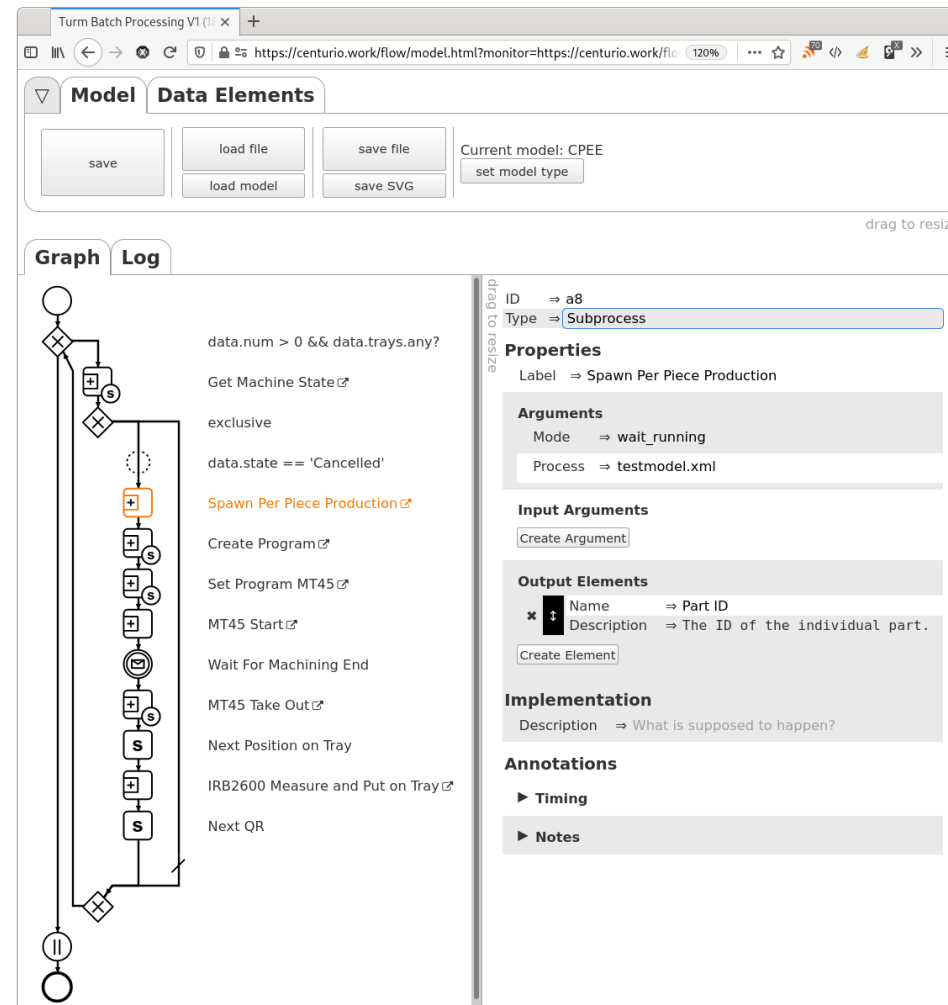


Non-interactive process automation

- Mix of legacy machines and hardware
- Goal: fully automatic production of batches
- What was automated?
 - Starting the machine (repetitive)
 - Taking manual measurements
- Benefits:
 - Skilled worker can work on other tasks
 - Reduction of errors by manual measurement

Conceptual design

- Requirements engineering
 - Textual description of input.
 - Textual description of output.
 - Limited set of task types.
- Specify control flow
- Batch View vs. Product View vs. Machine View
- Production vs. Production (Sensor) Data Collection



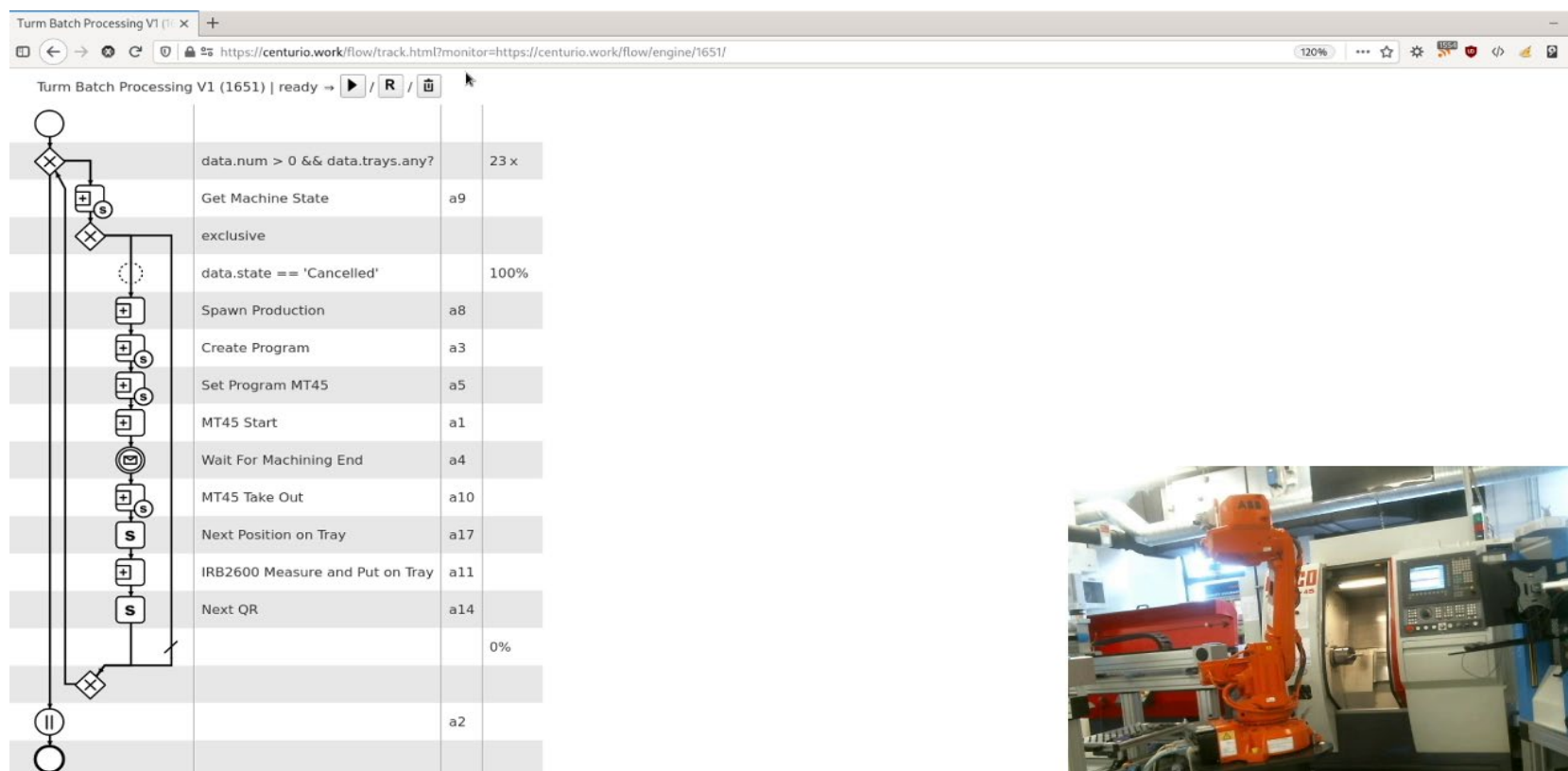
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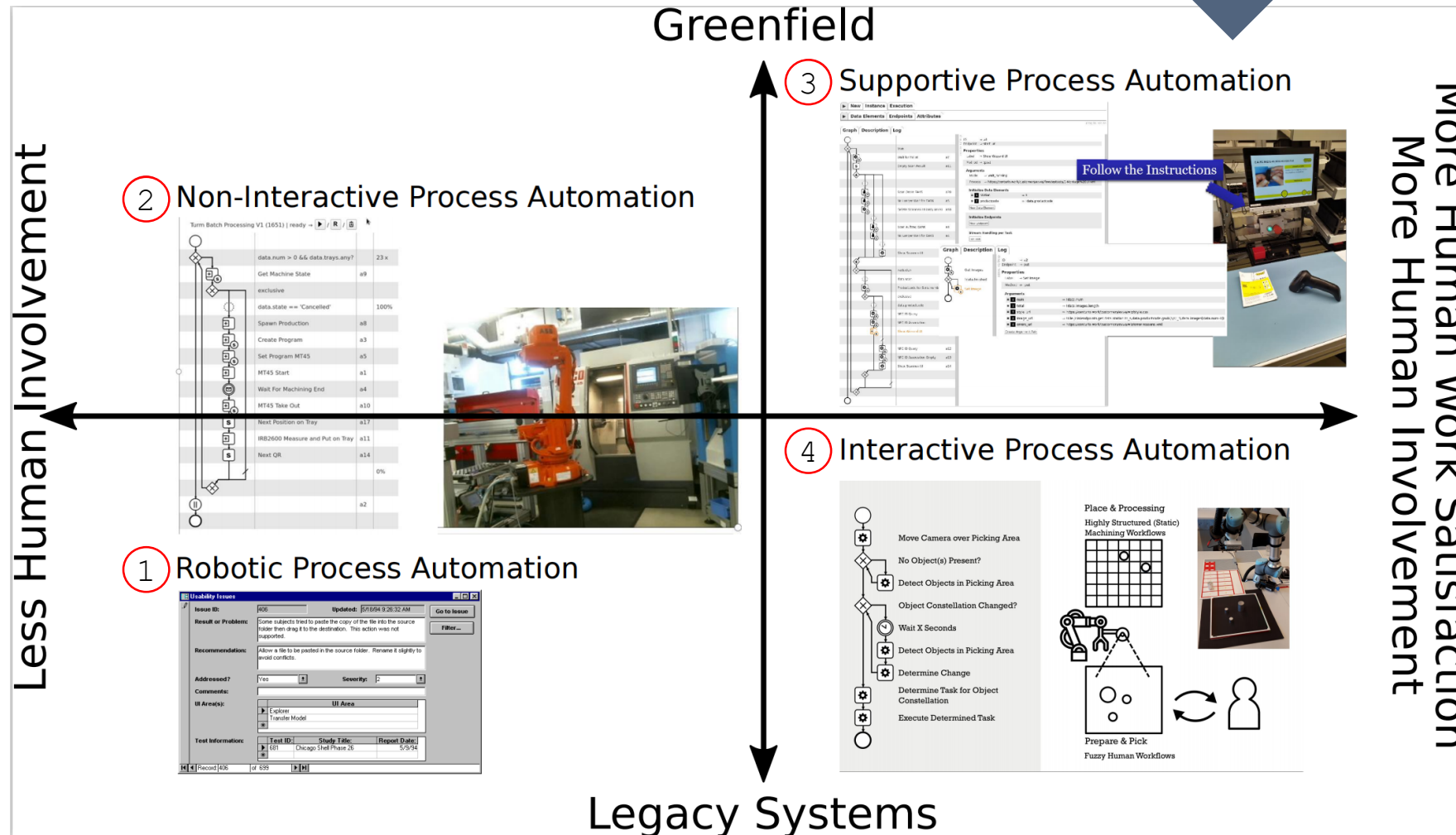
Execution design

- Bridge Execution Gap
 - Hook up functionalities & pass parameters
 - Handle response (Finalize)
 - Handle partial Response (Update)
 - Handle problems (Rescue)
 - Annotate for runtime compliance checking
 - Handle the collection of complementary sensor data, processing and automatic visualization (data stream handling)
- Step By Step Testing and Evolution

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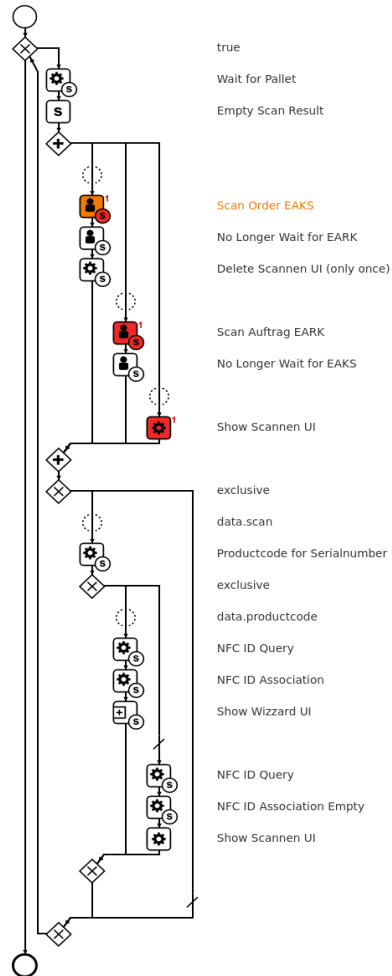
Process automation scenarios



Station 1: Scan Packing Label & Equipping Work Piece Carriers



EVVA E-Montage Station 1 (177) | running →



Station 1: Scan Packing Label & Equipping Work Piece Carriers



Follow the Instructions

Graph **Description** **Log**

ID	Description	Log
a1	Wait for Pallet	
a7	Empty Scan Result	
a11		
a20	Scan Order EAKS	
a5	No Longer Wait for EARK	
a10	Delete Scannen UI (only once)	
a3	Scan Auftrag EARK	
a4	No Longer Wait for EAKS	
a9	Show Scannen UI	
a8	Productcode for Serialnumber	
a2	NFC ID Query	
a6	NFC ID Association	
a1	Show Wizzard UI	
a12	NFC ID Query	
a13	NFC ID Association Empty	
a14	Show Scannen UI	

Properties

Label → Show Wizzard UI
Method → :post

Arguments

Mode → wait_running
Process → https://centurio.work/customers/evva/flow/testsets/E-Montage%20UI.xml

Initialize Data Elements

station → 1
productcode → !data.productcode

Initialize Endpoints

Stream Handling per Task

Data Stream Handling

Implementation

Annotations

Timing

Notes

Graph **Description** **Log**

ID	Description	Log
a2	Get Images	
!	!data.finished	
	Set Image	

Properties

Label → Set Image
Method → :put

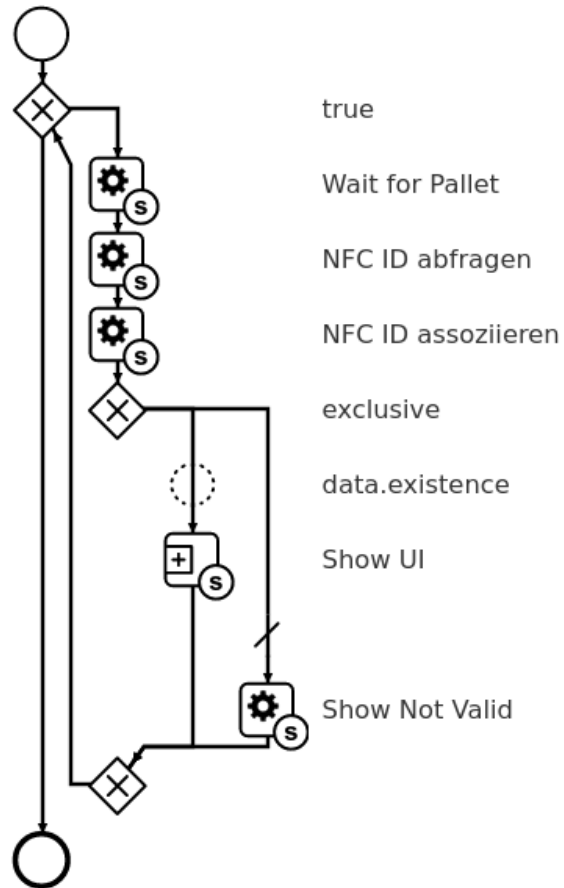
Arguments

num → !data.num
total → !data.images.length
style_url → https://centurio.work/customers/evva/wiz/style.css
image_url → !File.join(endpoints.get.data.station.to_s_data.productcode.gsub(/\/./, '_').data.images[data.num-1])
errors_url → https://centurio.work/customers/evva/wiz/error-reasons.xml

Station 2: Detect Work Piece Carrier, Show Associated Instructions



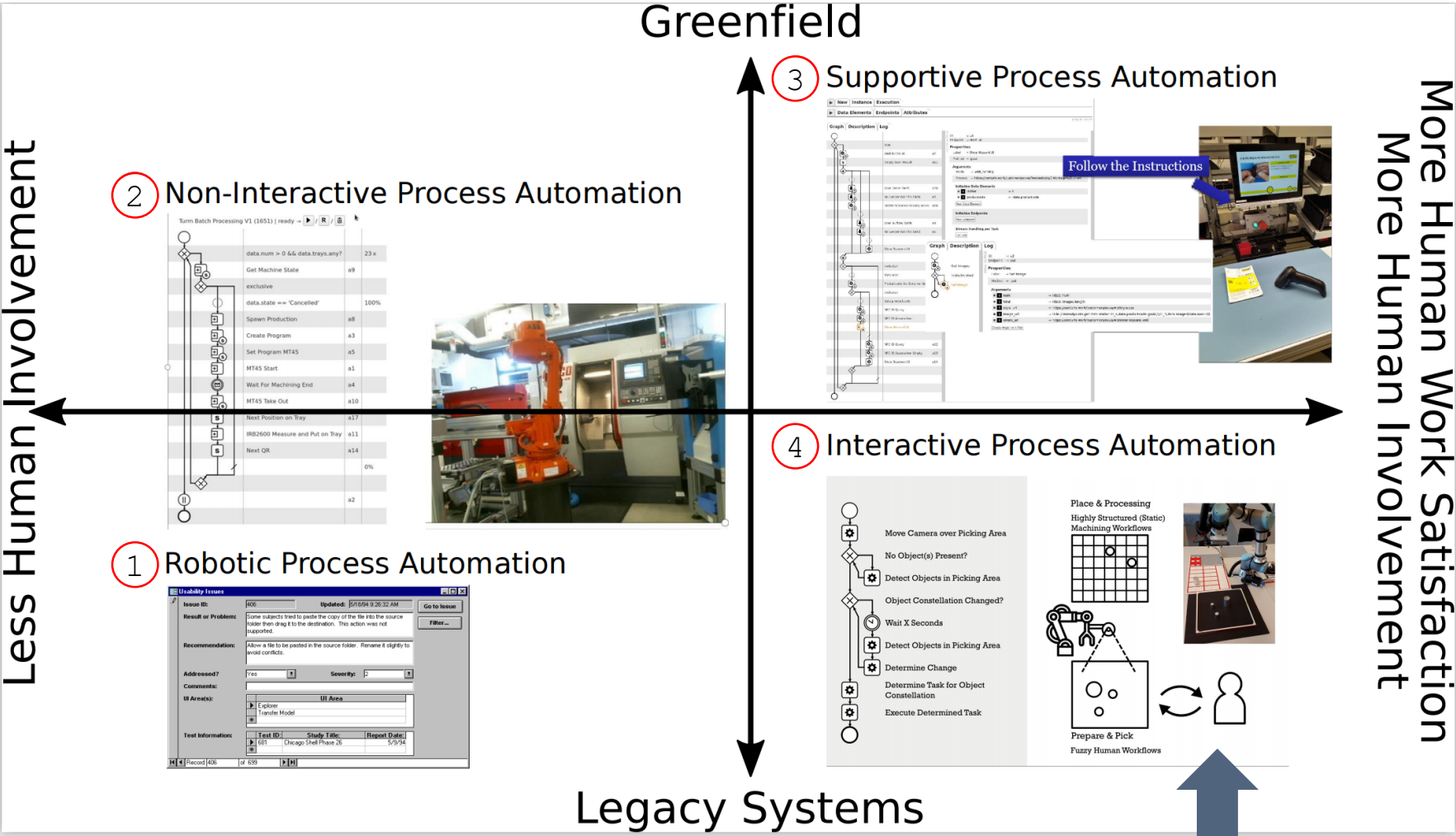
EVVA E-Montage Station 2 (960) | ready ⇒ ▶ / R / 🗑



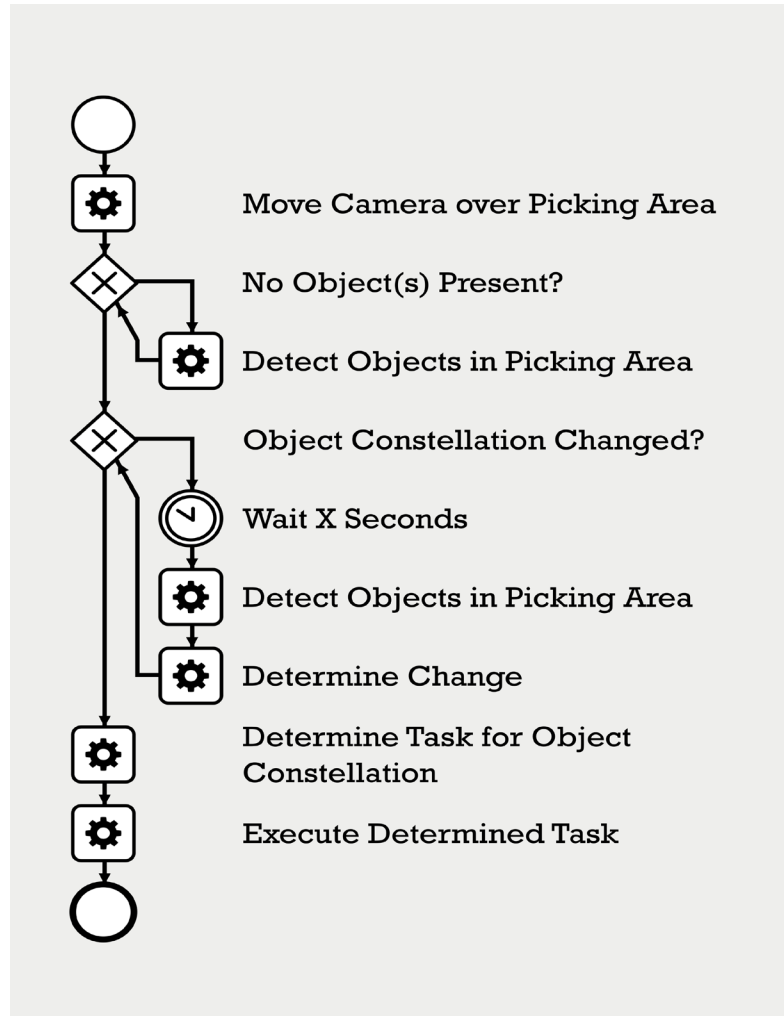
Follow the Instructions



Process automation scenarios

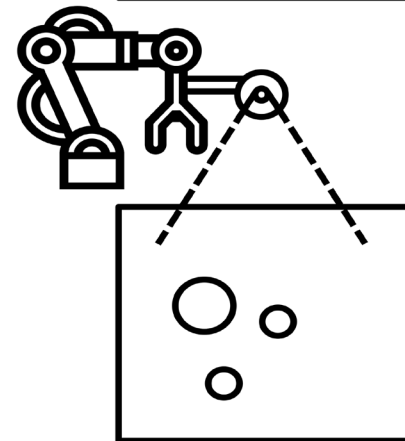
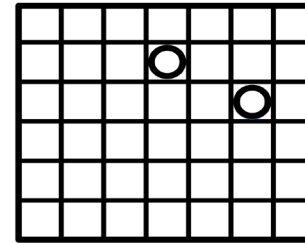


Interactive process automation



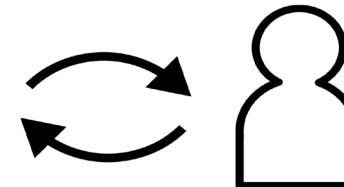
Place & Processing

Highly Structured (Static)
Machining Workflows



Prepare & Pick

Fuzzy Human Workflows

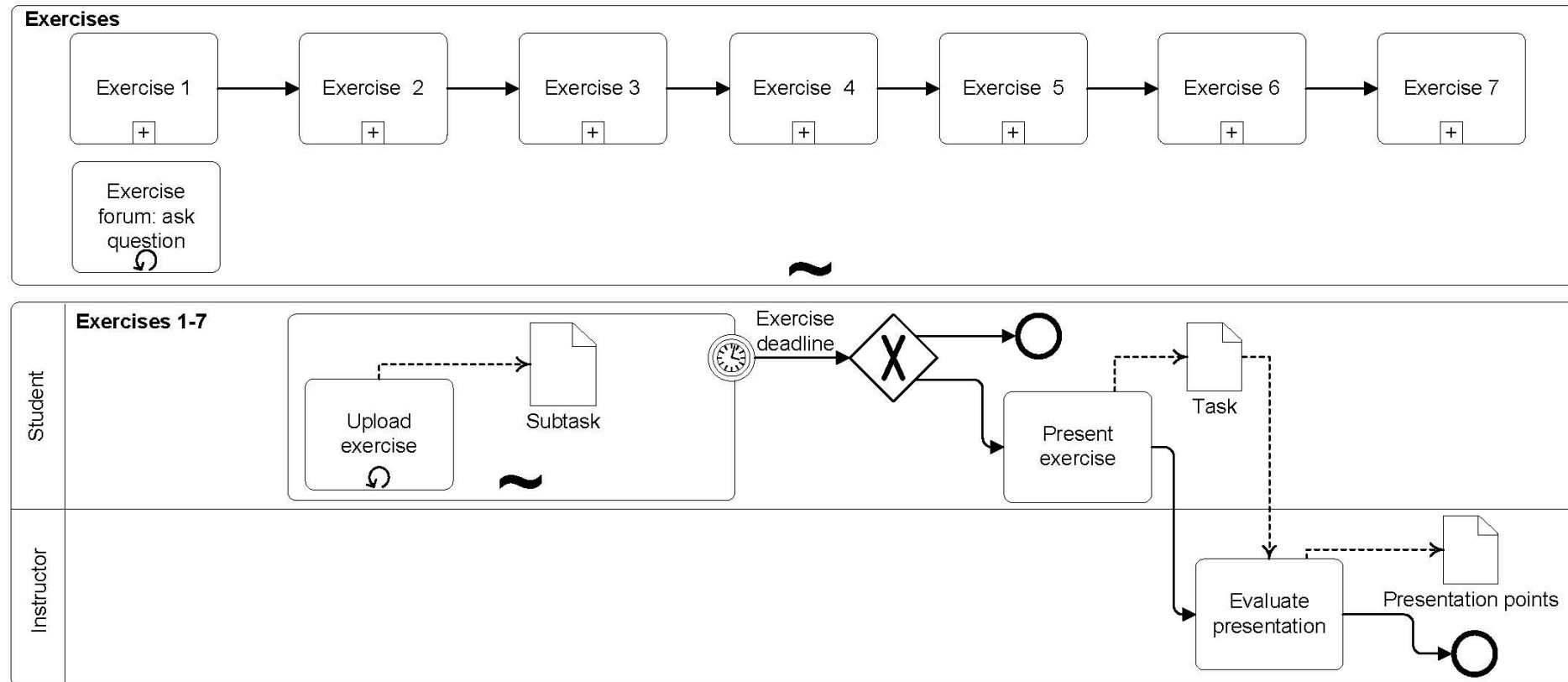


(c) A. Mangat

Read (open access): A. S. Mangat, J. Mangler, S. Rinderle-Ma:
Interactive Process Automation based on lightweight object
detection in manufacturing processes. Comput. Ind. 130: 103482
(2021)

Process technology and its application is domain-independent.

Higher education processes

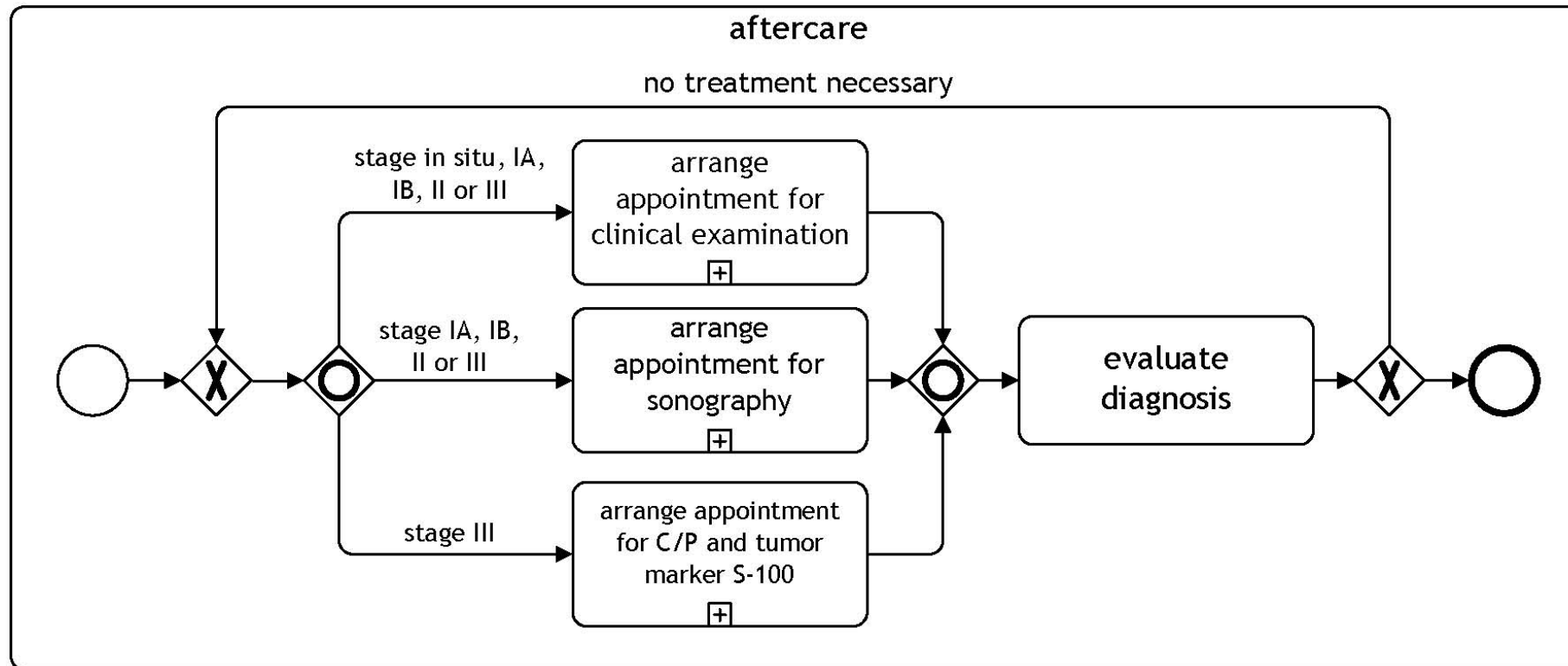


Source:

© Springer

Linh Thao Ly, Conrad Indiono, Jürgen Mangler, Stefanie Rinderle-Ma: Data Transformation and Semantic Log Purging for Process Mining. CAiSE 2012: 238-253

Skin cancer treatment processes



Source:
© Springer
Reinhold Dunkl, Karl Anton Fröschl, Wilfried Grossmann, Stefanie Rinderle-Ma: Assessing Medical Treatment Compliance Based on Formal Process Modeling. USAB 2011: 533-546

Process-aware Information Systems



- are aware of the logic of the implemented business process(es) (which is often defined in terms of an explicit *process model*)
- ensure that *process steps* (= *activities*) are executed in the right order during *runtime* (i.e., the PAIS manages the *flow of control*)
- control the *flow of data* between process activities (i.e., the data being exchanged between activities)
- provide an integrated view on *process and application data*
- know which *application component* shall be invoked when a particular activity is started
- pass input parameters to invoked applications and stores their output parameters in the *process data context*

Process-aware Information Systems



- add manual tasks (i.e., tasks requiring user interactions) to the *worklists* of those users who qualify for this task based on their assigned *roles* (e.g., secretary, medical doctor)
- *synchronize* concurrent changes of these worklists
- provide *reminders* if started activities are not finished in time
- enable *process participants* to *monitor* the progress of the executed processes and to record process-related events in a logfile (= *audit trail, execution log, event stream during runtime*)

Summary Chapter 1: Introduction

- Process technologies are key drivers for digital transformation
- Huge benefits through automation, integration, and mining
- Key skills:
 - process design and modeling, conceptual and execution
 - process-oriented application development
 - process analytics